

Samuel Bicalho Henriques

Materials Engineering Student — Crystallography and Materials Characterization
+33 7 61 41 04 32 • samuca.henriques@gmail.com • Metz, France • [LinkedIn](#) • [Lattes CV](#)

EDUCATION

B.Sc. Materials Engineering (5-year program)

In progress

Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Brazil

- Focus: crystallography, materials characterization, magnetic materials, phase transformations.

RESEARCH AND PROFESSIONAL EXPERIENCE

R&D Intern — Process Modelling and Development

May 2026 – Nov 2026

ArcelorMittal France, Maizières-lès-Metz — Global R&D

- Experimental characterization of a bench-scale gravity separation unit for mineral processing.
- Design of experiments, mass-balance analysis and statistical modelling of process data.
- Technical reporting and coordination of test campaigns with laboratory staff.

Undergraduate Researcher — Magnetic Materials (CAPES scholarship)

Aug 2025 – Jul 2026

CDTN — Nuclear Technology Development Center, Belo Horizonte

- Structural and magnetic study of rare-earth-doped nickel ferrite nanoparticles by XRD and Mössbauer spectroscopy.
- Rietveld refinement (FullProf) for phase quantification and crystallographic parameters.
- Designed and submitted a synchrotron beamtime proposal to CNPEM/Sirius, combining high-resolution XRD with maximum-entropy electron density mapping and XMCD.

Undergraduate Researcher — Biomaterials (CAPES scholarship)

Jan 2024 – Jan 2025

UFMG — Laboratory of Biomaterials

- Characterization of hybrid scaffolds by XRD and FTIR; comparative analysis of analytical methods for polymer characterization.

Extension Scholar — VALE Technical Expertise Program

Jan 2024 – Jan 2025

UFMG / VALE industrial partnership

- Coordinated and hosted a technical training program in mining and geometallurgy for VALE employees, managing scheduling, sessions and logistics.

Extension Scholar — Industrial Characterization Services

Jan 2023 – Dec 2023

Fundação Christiano Ottoni (FCO) / UFMG

- Performed commercial XRD and XRF analyses for external industrial clients; prepared certification reports.

PUBLICATIONS

Martins, T., Henriques, S. B., Reis, G. K. O., Amaral, M. M. S., Guimarães, M. F., Oliveira, G. L., Santos, A. O., Nunes, E. H. M. (2026). A Novel Green Method for Producing Medical-Grade Chitosan From Loligo plei Squid Pens. *Journal of Applied Polymer Science*, 143(25). DOI: 10.1002/app.70790

Martins, T., Oliveira, G. L., Henriques, S. B., Nunes, E. H. M. (2025). Comparative Analysis of Analytical Methods for Quantifying the Deacetylation Degree in Chitosan Polymer Chains. *Materials Research*, 28. DOI: 10.1590/1980-5373-MR-2025-0108

Mendonça, T. V., Henriques, S. B., et al. Characterization of Apatite Floatability. *ABM Week*, 2024.

Morais, F. R. M., Henriques, S. B., et al. Gold Recovery from Copper Waste. *ABM Week*, 2024.

TECHNICAL SKILLS

Characterization: XRD (Rietveld refinement), Mössbauer spectroscopy, XRF, FTIR, SEM/EDS.

Crystallography: FullProf Suite, HighScore, Profex, VESTA; electron density mapping, Fourier analysis.

Data and modelling: design of experiments (factorial DOE), regression modelling, mass balance analysis.

Software: OriginLab, LaTeX, Office Suite.

AWARDS AND ADDITIONAL TRAINING

Waelzholz Brasmetal Award 2025 — industrial excellence (ABM), co-recipient.

Bronze Medal, Minas Gerais Chemistry Olympiad (2019).

ThermoFisher Scientific: XRD & XRF equipment training (2024); HarvardX: Thermodynamics (2023); MITx: Nuclear Energy Systems (2022).

LANGUAGES AND EXTRACURRICULAR

Languages: Portuguese (native); English (C1 — CENEX/FALE-UFMG proficiency exam, 90/100, Oct 2025); French (in progress, currently living in France).

Student council member, Materials Engineering program, UFMG (2 years); Asset Director for 1 year.

Organizing team: Mining and Metallurgy Symposium, UFMG (2 editions); ore characterization event for industry, DEMIN/UFMG.